Name		Date of Performance	
Roll No.		Date of Submission	
Branch		Examined	

Title : Calibration of measuring instrument.

Instrument	**Specifications**
Dial Gauge	L.C. 0.01 mm

Introduction: The manufacturing tolerances in almost all the industries are becoming stringent due to increased awareness of quality. This also calls for high accuracy component in precision assembles and sub-assemblies. The quality control department is therefore is loaded with the periodic calibration of various measuring instruments. Since the accuracy of components depends largely on the accuracy of the measuring instruments like plunger type dial gauge, back plunger type dial gauges, lever type dial gauges and box gauges, periodic calibration is of vital quality. Assurance as well as cost reduction. The set of dial calibration tester enables us to test four different kinds of precision measuring instruments and all the required accessories are included in the set. The habit of periodic calibration has to be achieved right from the stage of technical education viz. engineering colleges, polytechnics and other institutions.

Why is periodic calibration required :-

- To grade a dial according to its accuracy and thereby to choose the application where it can be safely used.
- To determine worn out zone of travel facilitating dull utilization of dials.
- To inspect the dial after repairs and maintenance.
- To ascertain the exact point of discardation.

SCOPE: This procedure is to cover the dial calibration tester for the following range.

CALIBRATION EQUIPMENT:

- Slip gauge – 0 grade.

CALIBRATION PROCESS:

- Keep the dial on calibration testor on dial holding stand.

- Set the zero reading by touching the plunger to dial gauge platform.
- Give some small displacement by using knob on dial holding stand. Again adjust the zero with the pointer.
- Take 0.5 mm slip gauge. Keep it between dial pointer and dial stand base.
- See the readings. It should be 0.5 mm. if not then lose or tight the screw beneath the dust protection cap.
- Add 1 mm slip gauge. Repeat the process
- Add 2 mm slip gauge. Repeat the process
- Repeat this process until distance of 10 mm.
- Now remove, one by one each slip gauge and repeat above process
- See the readings, if any deviation from the readings, then lose or tight the screw, beneath the protection cap.

CONCLUSION: Hence, calibration of dial gauge is done and studied.

SUMMARY: calibration of instrument has been studied.

Question:

Q1) Enumerate the desirable qualities of Dial Gauges.

Q2) State the precaution to be taken while using Dial Gauge.

Q3) State the uses and practical application of Dial Gauge.